

# Cross-validation and hyperparameters

Give the data three roles, rotate the validation fold, and keep the test set for one final look.

## THE QUESTION

How do you choose a hyperparameter such as tree depth and still get an honest estimate of how the model will do?

## THE INTUITION

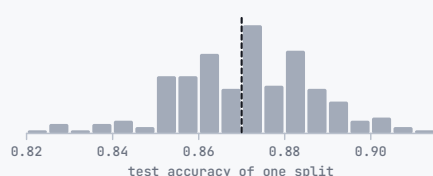
A score you used to make a choice stops being a fair judge. So hold out a test set, choose on validation data, and rotate the validation role over  $k$  folds so that one lucky split does not decide.

## KEY TAKEAWAY

Choose by the mean validation score over  $k$  folds, refit on all the training data, test once. To report a tuned score, nest the choice inside an outer CV.

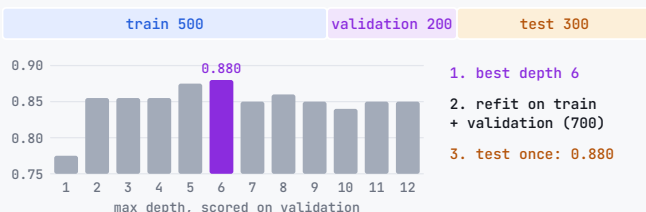
shuffle → hold out a test set →  $k$  folds on the rest → mean score per setting → refit with the best → test once

### 1 One split gives one noisy number



1000 points from two noisy moons (our re-creation of the notebook's data) and a tree with no depth limit. Only the random 70/30 split changes, yet test accuracy runs from 0.823 to 0.910, and 10 splits never test 21 of the points (28 expected, from  $1000 \cdot 0.7^{10}$ ). The notebook's 10 splits: 0.840 to 0.913.

### 2 Three roles: fit, choose, report



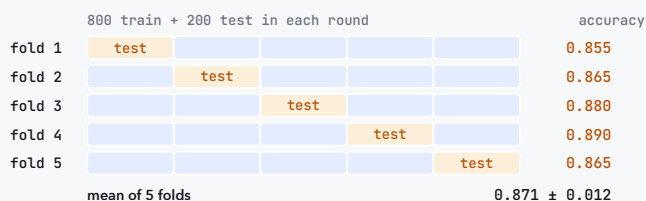
Training data fits the parameters, validation data chooses hyperparameters such as max depth, and the test set is used once, at the end. The slides' sketch: depths 1, 2 and 3 score 70%, 90% and 80% on validation, so keep depth 2. Here validation picks depth 6 (0.880); refit on all 700, it scores 0.880 on test. But the pick rests on one split of 200 points.

### 3 Shuffle before you split



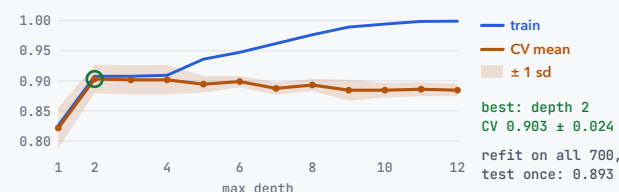
The slides' file has 80 Yes rows, then 20 No. Split it 80/20 in file order and the model never sees a No in training, then meets only No in the test. Order hurts SGD too: if point 2 always undoes point 1's step, the updates go round in a loop. Time series are the exception (panel 11).

### 4 k-fold: every point is tested once



Split the shuffled data into  $k$  equal folds. Round  $i$  trains on the other  $k - 1$  and tests on fold  $i$ : each point is tested once and trains  $k - 1$  times (100 points,  $k = 5$ : 80 per round). Report the mean  $\frac{1}{k} \sum_i acc_i$  and its sd; with equal folds it equals the pooled accuracy.

### 5 Pick the depth with the best mean CV score



5-fold CV on the 700 training points. Training accuracy keeps rising; the CV score peaks at depth 2 (0.903), just above depth 6 (0.899), the single validation set's pick. CV leaves  $k$  models and no final one: refit on all the training data with the chosen depth, then test once (0.893).

### 6 k-fold in NumPy

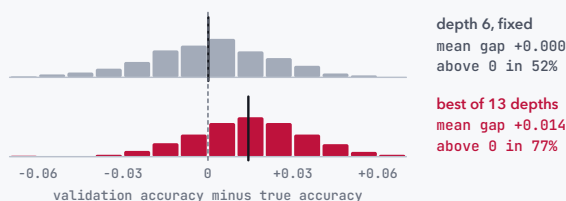
```
def kfold(n, k, seed=0):
    rng = np.random.default_rng(seed)
    idx = rng.permutation(n) # shuffle
    folds = np.array_split(idx, k)
    for i in range(k):
        rest = folds[:i] + folds[i+1:]
        yield np.concatenate(rest), folds[i]
def cv_score(fit, X, y, k=5):
    acc = []
    for tr, te in kfold(len(y), k):
        model = fit(X[tr], y[tr])
        pred = model(X[te])
        acc.append(np.mean(pred == y[te]))
    return np.mean(acc), np.std(acc)
```

fit returns a predict function. In scikit-learn: KFold and cross\_val\_score.

# Tune inside, test outside

Why the winning score flatters, how nested CV fixes it, and splits for rare classes, time order and preprocessing.

## 7 The winning validation score is optimistic



200 fresh datasets (500 train, 200 validation), true accuracy from 5000 new points. A depth fixed in advance is right on average. Picking the best of 13 depths also picks the luckiest noise: its score is 0.014 too high on average (0.901 against 0.886). The notebook's winner: validation 0.925, test 0.907.

## 8 Nested CV: choose inside, test outside

| outer loop: 5 folds of 1000 |      |      |      |      | depth | test  |
|-----------------------------|------|------|------|------|-------|-------|
| outer 1                     | test |      |      |      | 2     | 0.875 |
| outer 2                     |      | test |      |      | 5     | 0.890 |
| outer 3                     |      |      | test |      | 5     | 0.895 |
| outer 4                     |      |      |      | test | 2     | 0.890 |
| outer 5                     |      |      |      | test | 5     | 0.890 |

|         |            |            |            |                    |
|---------|------------|------------|------------|--------------------|
| inner 1 | validation | train      | train      | inside outer 1:    |
| inner 2 | train      | validation | train      | try depths 1 to 10 |
| inner 3 | train      | train      | validation | pick the best mean |

Each outer round holds out 200 points. On the other 800, a 3-fold CV scores depths 1 to 10; refit the best on the 800 and test on the 200, which never saw the choice. Fits:  $K_{out}(K_{in} | \text{grid}| + 1) = 155$ .

## 9 The chosen depth can differ by fold

| outer | d=1  | d=2  | d=3  | d=4  | d=5  | d=6  | d=7  | d=8  | d=9  | 10   | pick | test  |
|-------|------|------|------|------|------|------|------|------|------|------|------|-------|
| 1     | 80.1 | 89.4 | 89.4 | 89.3 | 89.4 | 88.1 | 88.4 | 88.1 | 88.1 | 87.9 | 2    | 0.875 |
| 2     | 80.4 | 88.6 | 88.6 | 88.2 | 89.6 | 89.4 | 88.7 | 88.0 | 88.4 | 88.1 | 5    | 0.890 |
| 3     | 80.0 | 88.1 | 88.1 | 88.1 | 88.7 | 88.1 | 87.5 | 87.1 | 88.0 | 86.9 | 5    | 0.895 |
| 4     | 81.0 | 89.9 | 89.9 | 89.8 | 88.7 | 88.0 | 88.9 | 87.1 | 87.1 | 88.3 | 2    | 0.890 |
| 5     | 80.1 | 89.6 | 89.6 | 89.7 | 89.9 | 89.2 | 89.1 | 88.7 | 87.8 | 87.8 | 5    | 0.890 |
| mean  | 80.3 | 89.1 | 89.1 | 89.0 | 89.3 | 88.6 | 88.5 | 87.8 | 87.9 | 87.8 |      | 0.888 |

inner 3-fold accuracy (%) on the 800 outer-training points

Depths 2 and 5 both win, so nested CV scores the recipe (tune, then fit), not one tree. Report the outer  $0.888 \pm 0.007$ ; plain CV on all 1000 points gives its winner 0.893, which is optimistic. To ship, rerun the inner CV on all the data and refit.

## 13 Pause and think

Report the best fold? No, it is a lucky split; report the mean and sd.

0.81 accuracy on 9:1 data?

Guessing at the class rates gets  $0.9^2 + 0.1^2 = 0.82$  (the dummy-baselines notebook: 0.81).

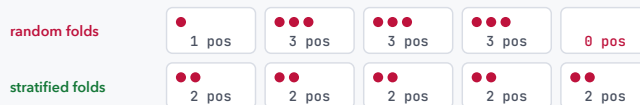
Which model do you ship? One refit on all the training data with the chosen setting.

Tune depth on the test set? Then it is a validation score, and it flatters (panel 7).

Leave-one-out, 1000 points, 10 depths? 10,000 fits; 5-fold needs 50.

Does nested CV score the final model? It scores the recipe: tune, then fit.

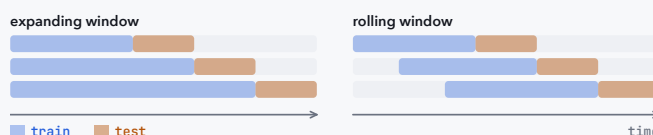
## 10 Rare classes need stratified folds



20,000 random 5-fold splits: 43% leave a fold with no positives

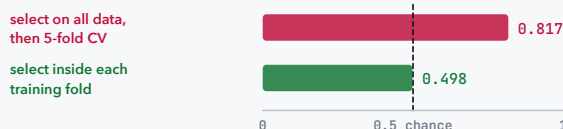
10 positives in 100 points: one random 5-fold split left a fold with none, where recall is undefined. Stratified folds keep the class ratio. Leave-one-out ( $k = n$ ) suits tiny data:  $n$  fits and no randomness, but slow, with a high-variance estimate.

## 11 In time order, train on the past



Random folds assume independent points; in a series they let the model peek at the future. A 200-step random walk, predicted from the nearest training time: random 5-fold MSE 0.86, forward chaining 4.14. Expanding windows keep all the past; rolling windows a fixed length.

## 12 Anything fitted to data goes inside the fold



Scaling, imputation and feature selection learn from data, so fit them on the training folds only and apply them to the test fold. The slides' example: a mean and sd for scaling, computed before splitting. Our test: 100 points, random labels, 2000 noise features; keep the 10 most label-correlated, nearest-centroid classifier, 30 repeats. Selecting on all data scores 0.817; inside each fold, 0.498. Keep each group (one patient) in one fold.

shuffle, then split · choose on validation folds · refit on all training data · test once · nest to report a tuned score

Before: the decision trees II and bias and variance sheets; next: the ensembles sheet. Source: lecture slides "Cross-validation", "Bias-variance and cross validation" and "Shuffling", and the hyperparameter-tuning and dummy-baselines notebooks (nipunbatra.github.io/ml-teaching). Data: our seeded re-creation of the notebook's two moons; notebook numbers are quoted.